MATH 231 – Linear Algebra I

The Matrix Unit, Part 2 ▪ Column Space, Row Space, Rank, and the 2×2 Determinant

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About this document. This is Part 2 of the matrix unit. Section numbering runs continuously through the parts of the unit, so a reference such as §2.5 points into Part 1 and means exactly what it says. References into the vector unit always name it.

Part	Sections	Subject
1	§1–§9	linear systems, the matrix, and its arithmetic
2	§10–§14	column space, row space, rank, the 2×2 determinant
3	§15–§21	matrices that move the plane; orthogonal matrices
4	§22–§25	the determinant as an area; rotations
5	§26–§29	projection matrices and the direct sum
6	§30–§34	the null space, rank–nullity, the 2×2 inverse
7	§35–§39	the determinant in general
8	§40–§45	solving linear systems by elimination
9	§46–§49	least squares
10	§50–§54	Cramer's rule, the inverse, the pseudo-inverse
11	§55–§58	Gram–Schmidt and the QR decomposition

Assumed. From Part 1: matrices and their shapes (§3), the matrix–vector product read by rows (§2.3) and by columns (§2.5), the transpose (§6), and the algebraic laws M1–M6 of matrix multiplication (§8.4). From the vector unit: span (§16.3), dimension (§17), linear independence (§18.4) and the test $D = u_1v_2 - u_2v_1$ (§18.5), subspaces (§19), and $\mathbb{R}^n$ (§24).

What you should be able to do afterwards. Describe the column space and the row space of a matrix and say which space each lives in; decide whether $\mathbf{Ax} = \mathbf{b}$ has a solution by asking whether $\mathbf{b}$ lies in the column space; find the rank of a small matrix by inspection; read a matrix as a function that moves vectors, and build the matrix of such a function from the images of $\mathbf{e}_1, \ldots, \mathbf{e}_n$; compute a 2×2 determinant and say what its size, its sign, and its vanishing mean for the way the matrix moves the plane.

Part 1 ended with a single equation, $\mathbf{Vx} = \mathbf{u}$, that holds all n equations of a linear system at once. It ended with a promise too: a method for solving it. That method, Gaussian elimination, is coming, in Part 8. Before learning *how* to find a solution, it pays to ask *whether* there is one. Problem P5 of §2.6 produced a system with no solution at all, and nothing about the notation $\mathbf{Vx} = \mathbf{u}$ warned us. This part builds the language that answers the question exactly: the *column space* of a matrix decides whether a system can be solved, the *rank* measures how much of the space the columns reach, and for a 2×2 matrix a single number, the *determinant*, settles both.

10. The column space

10.1. Everything $\mathbf{Ax}$ can be

Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ have columns $\mathbf{a}_1, \dots, \mathbf{a}_n$. Each column has one entry per row, so each column is a vector in $\mathbb{R}^m$. The column reading of a matrix–vector product (§2.5) says

$$\mathbf{Ax} \;=\; x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

So as $\mathbf{x}$ ranges over all of $\mathbb{R}^n$, the product $\mathbf{Ax}$ ranges over every linear combination of the columns, and over nothing else. The collection of all linear combinations has a name already: it is the span (the vector unit, §16.3).

Definition 10.1 (Column space)**.** The **column space** of $\mathbf{A} \in \mathbb{R}^{m\times n}$ is the set of all products $\mathbf{Ax}$:

$$\operatorname{col}[\mathbf{A}] \;=\; \{\, \mathbf{Ax} : \mathbf{x} \in \mathbb{R}^n \,\} \;=\; \operatorname{span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\}.$$

It is a set of vectors in $\mathbb{R}^m$.

The two descriptions are the same set, by the column reading. The first says what the column space is *for*: it is everything the matrix can produce. The second says how to *see* it: look at the columns and ask what they span.

10.2. It is a subspace

From the vector unit. A subspace of $\mathbb{R}^m$ is a set of vectors that contains $\mathbf{0}$ and is closed under addition and scalar multiplication (§19.3), and every span is automatically one (§19.2). So the second description already tells us that $\operatorname{col}[\mathbf{A}]$ is a subspace. The first description gives a proof written in matrix language, which is worth seeing because the same three lines will be reused for other subspaces in Part 6.

Theorem 10.1. *For every $\mathbf{A} \in \mathbb{R}^{m\times n}$, the column space $\operatorname{col}[\mathbf{A}]$ is a subspace of $\mathbb{R}^m$.*

Proof. It contains $\mathbf{0} = \mathbf{A0}$. If $\mathbf{Ax}$ and $\mathbf{Ay}$ are in it, so is their sum, because $\mathbf{Ax} + \mathbf{Ay} = \mathbf{A}(\mathbf{x}+\mathbf{y})$ by distributivity (M2 of §8.4). And if c is a scalar, $c(\mathbf{Ax}) = \mathbf{A}(c\mathbf{x})$ by M4, which is again in it. □

10.3. When $\mathbf{Ax} = \mathbf{b}$ has a solution

The definition makes the following statement almost a tautology, and it is still the most useful sentence in this part.

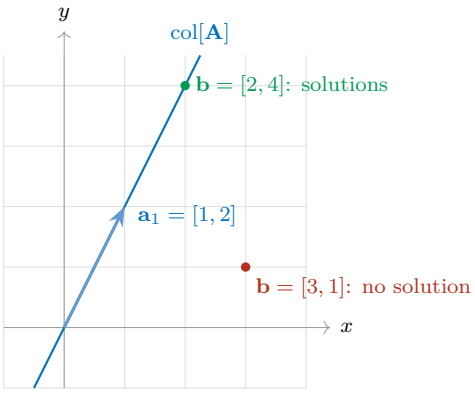
Theorem 10.2 (Solvability)**.** *The system $\mathbf{Ax} = \mathbf{b}$ has at least one solution if and only if $\mathbf{b} \in \operatorname{col}[\mathbf{A}]$.*

Proof. A solution is a vector $\mathbf{x}$ with $\mathbf{Ax} = \mathbf{b}$, and $\operatorname{col}[\mathbf{A}]$ is by definition the set of vectors of the form $\mathbf{Ax}$. □

Example 10.1 (§2.6, Problem P5, revisited)**.** Problem P5 asked for the coordinates of $\mathbf{u} = [\,3, 1\,]$ against $\mathbf{v}_1 = [\,1, 2\,]$ and $\mathbf{v}_2 = [\,2, 4\,]$, that is, for a solution of

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

The second column is twice the first, so the column space is the span of $[\,1, 2\,]$ alone: the line $y = 2x$ through the origin. The point $[\,3, 1\,]$ is not on that line, so there is no solution, which is what P5 found by algebra. Change the right-hand side to $\mathbf{b} = [\,2, 4\,]$, which *is* on the line, and solutions appear: $x_1 + 2x_2 = 2$ is satisfied by $[\,2, 0\,]$, by $[\,0, 1\,]$, and by infinitely many other pairs.



The vector unit's review sorted the outcomes of a 2×2 system into three cases: one solution, none, or infinitely many. The first half of that sorting is now a statement about any system of any size: *no solution* means exactly that $\mathbf{b}$ lies outside col[$\mathbf{A}$]. The other half, one solution versus infinitely many, is decided by a second subspace, which arrives in Part 6.

10.4. Examples

Example 10.2. Five column spaces, read off the columns.

(a) $\mathbf{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$: the columns $\mathbf{e}_1, \mathbf{e}_2$ span $\mathbb{R}^2$, so col[$\mathbf{I}$] $= \mathbb{R}^2$, and $\mathbf{Ix} = \mathbf{b}$ is solvable for every $\mathbf{b}$ (by $\mathbf{x} = \mathbf{b}$).

(b) $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$: the line through $[\,1, 2\,]$, as in Example 10.1.

(c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \in \mathbb{R}^{3\times 2}$: the columns are $\mathbf{e}_1, \mathbf{e}_2$ of $\mathbb{R}^3$, so the column space is the xy-plane, $\{[\,x, y, 0\,]\}$. The system with $\mathbf{b} = [\,1, 2, 0\,]$ is solvable; with $\mathbf{b} = [\,1, 2, 3\,]$ it is not, since nothing the matrix produces has a nonzero third entry.

(d) $\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \end{bmatrix} \in \mathbb{R}^{2\times 3}$: the first two columns already span $\mathbb{R}^2$ (they are independent, by the test of the vector unit's §18.5: $D = 1 \cdot 1 - 0 \cdot 1 = 1 \neq 0$), so the column space is all of $\mathbb{R}^2$. The third column adds nothing new.

(e) The 2×2 zero matrix: every product is $\mathbf{0}$, so the column space is $\{\mathbf{0}\}$.

Remark. Part (d) shows something worth stating once. A column that is a linear combination of the others does not enlarge the column space: it is already in the span of the rest. So a matrix can have many columns and a small column space, and the number that matters is not how many columns there are but how many *independent directions* they contain. That number is the rank, in §12.

11. The row space

11.1. The definition

A matrix has rows as well as columns, and the rows span something too. Row i of $\mathbf{A} \in \mathbb{R}^{m\times n}$ has one entry per column, so the rows $\mathbf{r}_1, \ldots, \mathbf{r}_m$ are vectors in $\mathbb{R}^n$.

Definition 11.1 (Row space)**.** The **row space** of $\mathbf{A} \in \mathbb{R}^{m\times n}$ is the span of its rows:

$$\text{row}[\mathbf{A}] \;=\; \text{span}\{\mathbf{r}_1, \ldots, \mathbf{r}_m\}.$$

It is a set of vectors in $\mathbb{R}^n$.

Transposing turns rows into columns (§6.3), so the rows of $\mathbf{A}$ are exactly the columns of $\mathbf{A}^\mathsf{T}$, and

$$\text{row}[\mathbf{A}] \;=\; \text{col}[\mathbf{A}^\mathsf{T}].$$

That one line gives everything §10 proved, for free: the row space is a subspace of $\mathbb{R}^n$, because it is the column space of the $n \times m$ matrix $\mathbf{A}^\mathsf{T}$.

11.2. Two different homes

For a matrix that is not square, the column space and the row space are not even in the same space.

Example 11.1. Let $\mathbf{A} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix} \in \mathbb{R}^{2\times 3}$. The columns $[1, 0]$, $[0, 1]$, $[2, 3]$ are vectors in $\mathbb{R}^2$, and the first two already span it, so $\text{col}[\mathbf{A}] = \mathbb{R}^2$. The rows $[1, 0, 2]$ and $[0, 1, 3]$ are vectors in $\mathbb{R}^3$. Neither is a multiple of the other, so they span a plane through the origin in $\mathbb{R}^3$. One subspace is the whole of $\mathbb{R}^2$; the other is a plane inside $\mathbb{R}^3$.

A common confusion (Column space and row space are different subspaces). Even for a square matrix, where both live in $\mathbb{R}^n$, the two are usually different. Take $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$. Its columns $[1, 0]$ and $[2, 0]$ span the x-axis, so $\text{col}[\mathbf{A}]$ is the x-axis. Its rows are $[1, 2]$ and $[0, 0]$, so $\text{row}[\mathbf{A}]$ is the line through $[1, 2]$. Two different lines. They coincide when $\mathbf{A}$ is symmetric, since then $\mathbf{A}^\mathsf{T} = \mathbf{A}$; that is why Example 10.1's matrix, which is symmetric, has the same line for both.

11.3. What the row space does

The column space is what the column reading of $\mathbf{Ax}$ produces. The row space belongs to the *row* reading (§2.3): the i-th entry of $\mathbf{Ax}$ is the dot product of row i with $\mathbf{x}$,

$$(\mathbf{Ax})_i \;=\; \langle \mathbf{r}_i, \mathbf{x} \rangle.$$

So the rows are the directions along which the matrix *measures* its input. A vector orthogonal to every row gets measured as zero in every slot, so it is sent to $\mathbf{0}$. Those vectors form a subspace of their own, the null space, and §31 shows that it is exactly the set of vectors orthogonal to the row space.

12. Rank

12.1. The definition

From the vector unit. The dimension of a subspace is the number of vectors in any basis for it (§17, and §24 in $\mathbb{R}^n$). A line through the origin has dimension 1, a plane through the origin dimension 2, the subspace $\{\mathbf{0}\}$ dimension 0.

Definition 12.1 (Rank)**.** The **rank** of $\mathbf{A} \in \mathbb{R}^{m\times n}$ is the dimension of its column space:

$$\text{rank}(\mathbf{A}) \;=\; \dim \text{col}[\mathbf{A}].$$

12.2. Rank counts independent columns

A basis for the column space can always be chosen from among the columns themselves. Start with all n columns. If they are dependent, one of them is a combination of the others (the vector unit, §18.4); delete it, and by the remark in §10.4 the span does not change. Repeat until what remains is independent. What remains still spans col[$\mathbf{A}$] and is independent, so it is a basis. Hence:

> rank($\mathbf{A}$) is the largest number of linearly independent columns of $\mathbf{A}$.

Two bounds follow at once. There are only n columns, so $\operatorname{rank}(\mathbf{A}) \le n$. And col[$\mathbf{A}$] is a subspace of $\mathbb{R}^m$, which cannot hold more than m independent vectors, so $\operatorname{rank}(\mathbf{A}) \le m$. Together,

$$\operatorname{rank}(\mathbf{A}) \;\le\; \min(m, n).$$

12.3. Examples, by inspection

For small matrices the rank can be read off without computation, by spotting which columns are combinations of which.

Example 12.1.

(a) Any zero matrix has rank 0: its column space is $\{\mathbf{0}\}$.

(b) $\mathbf{I}_n$ has rank n: its columns $\mathbf{e}_1, \dots, \mathbf{e}_n$ are independent.

(c) $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ has rank 1: the second column is twice the first, and the first is not zero.

(d) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ has rank 2: the columns $[\,1, 3\,]$ and $[\,2, 4\,]$ are not parallel ($D = 4 - 6 = -2 \neq 0$).

(e) $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$ has rank 2: the third column is the sum of the first two, and the first two are not parallel.

(f) The 3×2 matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$ has rank $2 = \min(3, 2)$.

(g) An outer product has rank 1. With $\mathbf{u} = [\,1, 2, 3\,]$ and $\mathbf{v} = [\,1, -1\,]$,

$$\mathbf{u}\mathbf{v}^{\mathsf{T}} = \begin{bmatrix} 1 & -1 \\ 2 & -2 \\ 3 & -3 \end{bmatrix},$$

and in general column j of $\mathbf{u}\mathbf{v}^{\mathsf{T}}$ is $v_j\mathbf{u}$ (§7.2). Every column is a multiple of $\mathbf{u}$, so when $\mathbf{u}$ and $\mathbf{v}$ are nonzero the column space is the line through $\mathbf{u}$.

12.4. Row rank

The rows could be counted the same way: $\dim \operatorname{row}[\mathbf{A}]$ is the largest number of independent rows. Nothing in the definitions suggests that this should equal the rank. It does.

For a 2×2 matrix $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ it can be checked directly. The vector unit's test (§18.5) says the columns $[\,a, c\,]$ and $[\,b, d\,]$ are independent exactly when

$$D_{\text{cols}} \;=\; a \cdot d - c \cdot b \;\neq\; 0,$$

and the rows $[\,a, b\,]$ and $[\,c, d\,]$ are independent exactly when

$$D_{\text{rows}} \;=\; a \cdot d - b \cdot c \;\neq\; 0.$$

These are the same number. So the columns are independent exactly when the rows are, and both have rank 2 together. If they are dependent, the matrix has rank 1 both ways unless it is the zero matrix, which has rank 0 both ways.

Theorem 12.1 (Row rank equals column rank)**.** *For every* $\mathbf{A} \in \mathbb{R}^{m \times n}$*,* $\dim \operatorname{row}[\mathbf{A}] = \dim \operatorname{col}[\mathbf{A}] = \operatorname{rank}(\mathbf{A})$.

The 2×2 case was just checked. The general proof needs the null space and is given in §33.3.

Remark. It is a stronger statement than it looks. A 2×100 matrix has one hundred columns in $\mathbb{R}^2$ and two rows in $\mathbb{R}^{100}$. The theorem says that however the entries are chosen, the hundred columns span exactly as many dimensions as the two rows do. Since there are only two rows, that is at most two, which is also clear from the columns' side, since they live in $\mathbb{R}^2$. The theorem is that the two counts agree *exactly*, not just in their upper bound.

13. A matrix moves the plane

The column space is a static object: a set of vectors. This section adopts the point of view that makes it dynamic. A matrix *does* something to a vector: it takes it in and hands back another.

13.1. A matrix as a function

Fix $\mathbf{A} \in \mathbb{R}^{m \times n}$. For each input $\mathbf{x} \in \mathbb{R}^n$ the product $\mathbf{A}\mathbf{x}$ is a vector in $\mathbb{R}^m$. So $\mathbf{A}$ defines a function

$$\mathbf{x} \;\longmapsto\; \mathbf{A}\mathbf{x}, \qquad \mathbb{R}^n \longrightarrow \mathbb{R}^m,$$

with inputs in $\mathbb{R}^n$ and outputs in $\mathbb{R}^m$. Its set of outputs is $\operatorname{col}[\mathbf{A}]$. When $m = n = 2$, the inputs and outputs are both points of the plane, and we can watch the matrix pick each point up and set it down somewhere else.

13.2. Linearity

Two of the laws in §8.4, M2 and M4, become statements about this function once the vectors are read as $n \times 1$ matrices.

Theorem 13.1. *For every* $\mathbf{A} \in \mathbb{R}^{m \times n}$*, all* $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ *and every scalar* c*,*

$$\mathbf{A}(\mathbf{x} + \mathbf{y}) = \mathbf{A}\mathbf{x} + \mathbf{A}\mathbf{y}, \qquad \mathbf{A}(c\mathbf{x}) = c\,(\mathbf{A}\mathbf{x}).$$

These two properties have a name.

Definition 13.1 (Linear transformation)**.** A function $T : \mathbb{R}^n \to \mathbb{R}^m$ is **linear**, or a **linear transformation**, if

$$T(\mathbf{x}+\mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y}) \qquad \text{and} \qquad T(c\mathbf{x}) = cT(\mathbf{x})$$

for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and all scalars c.

So every matrix gives a linear transformation. One consequence is immediate and will matter in Part 3: taking $c = 0$ gives $T(\mathbf{0}) = \mathbf{0}$. *A linear transformation never moves the origin.*

13.3. Column j is where $\mathbf{e}_j$ goes

Feed the standard basis vector $\mathbf{e}_j$ into $\mathbf{A}$. By the column reading, $\mathbf{A}\mathbf{e}_j = 0\mathbf{a}_1 + \cdots + 1\mathbf{a}_j + \cdots + 0\mathbf{a}_n = \mathbf{a}_j$.

> Column j of $\mathbf{A}$ is $\mathbf{A}\mathbf{e}_j$, the place where $\mathbf{A}$ sends the j-th standard basis vector.

The converse is what makes this useful. Suppose $T : \mathbb{R}^n \to \mathbb{R}^m$ is *any* linear transformation, described in words or by a formula, with no matrix in sight. Write the input in the standard basis, $\mathbf{x} = x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n$, and apply linearity:

$$T(\mathbf{x}) = x_1T(\mathbf{e}_1) + \cdots + x_nT(\mathbf{e}_n) = \begin{bmatrix} T(\mathbf{e}_1) & \cdots & T(\mathbf{e}_n) \end{bmatrix} \mathbf{x},$$

the last step being the column reading again. So T is multiplication by the matrix whose columns are $T(\mathbf{e}_1), \ldots, T(\mathbf{e}_n)$.

Theorem 13.2. *A linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is completely determined by the n vectors $T(\mathbf{e}_1), \ldots, T(\mathbf{e}_n)$. It is multiplication by the matrix whose j-th column is $T(\mathbf{e}_j)$.*

Example 13.1. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be $T([\,x, y\,]) = [\,2x,\ x + y\,]$. It is linear: each output entry is a combination of x and y with fixed coefficients. $T(\mathbf{e}_1) = T([\,1, 0\,]) = [\,2, 1\,]$ and $T(\mathbf{e}_2) = T([\,0, 1\,]) = [\,0, 1\,]$, so

$$T \text{ is multiplication by } \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}, \qquad \text{and indeed} \qquad \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ x+y \end{bmatrix}. \quad \checkmark$$

This is the recipe Part 3 runs on. To build the matrix of a geometric motion of the plane, decide where the motion sends $\mathbf{e}_1$ and $\mathbf{e}_2$, and write those two vectors down as columns.

Remark. Linear transformations between spaces other than $\mathbb{R}^n$, and the reason matrix multiplication is defined the way §8.1 defined it, are taken up near the end of the course. Here a matrix acting on $\mathbb{R}^n$ is all we need.

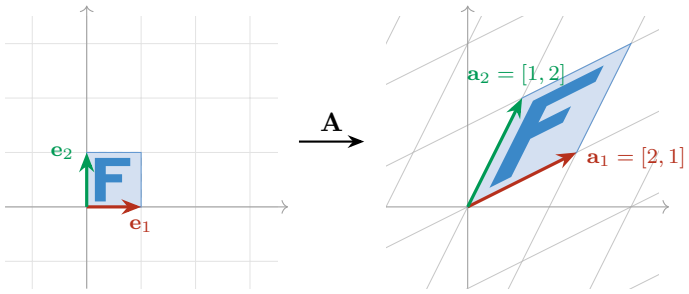
13.4. What happens to the grid

In the plane, the unit square has corners $\mathbf{0}$, $\mathbf{e}_1$, $\mathbf{e}_2$ and $\mathbf{e}_1 + \mathbf{e}_2$. A 2×2 matrix sends them to $\mathbf{0}$, $\mathbf{a}_1$, $\mathbf{a}_2$ and $\mathbf{a}_1 + \mathbf{a}_2$, and by linearity it sends every point $x_1\mathbf{e}_1 + x_2\mathbf{e}_2$ of the square to the point $x_1\mathbf{a}_1 + x_2\mathbf{a}_2$. So:

- the unit square goes to the parallelogram with sides $\mathbf{a}_1$ and $\mathbf{a}_2$;
- the whole coordinate grid goes to a grid of parallelograms, its lines running along $\mathbf{a}_1$ and $\mathbf{a}_2$, still parallel and still evenly spaced;

- the origin stays where it is, and lines through the origin go to lines through the origin (or, if the matrix crushes them, to the origin itself).

The figure shows $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. The letter F drawn in the unit square is carried along with it, so that the motion of the whole plane can be seen in one shape.



before after

14. The 2×2 determinant

14.1. The number D, named

From the vector unit. The vector unit's §18.5 found a single number that decides whether two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ span the plane: $D = u_1v_2 - u_2v_1$, nonzero exactly when they do. It closed with the promise that such a number "is worth having a name, and it has one." Here is the name. Put $\mathbf{u}$ and $\mathbf{v}$ in as the columns of a matrix.

Definition 14.1 (Determinant of a 2×2 matrix)**.** The **determinant** of $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is the number

$$\det(\mathbf{A}) \;=\; ad - bc.$$

"Down the main diagonal, minus up the other diagonal." With columns $[a, c]$ and $[b, d]$ this is exactly the vector unit's D.

Example 14.1.

$$\det \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = 4 - 1 = 3, \qquad \det \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = 4 - 4 = 0, \qquad \det \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = 0 - 1 = -1, \qquad \det \mathbf{I} = 1.$$

14.2. The determinant decides the rank

Everything §10–§12 established for a 2×2 matrix now collapses into one test.

Theorem 14.1. *For $\mathbf{A} \in \mathbb{R}^{2\times2}$ the following are equivalent:*

(i) $\det(\mathbf{A}) \neq 0$*;*

(ii) the columns of $\mathbf{A}$ *are linearly independent;*

(iii) $\operatorname{rank}(\mathbf{A}) = 2$*;*

(iv) $\operatorname{col}[\mathbf{A}] = \mathbb{R}^2$*;*

(v) $\mathbf{Ax} = \mathbf{b}$ *has a solution for every* $\mathbf{b} \in \mathbb{R}^2$*.*

Proof. (i) $\Longleftrightarrow$ (ii) is the vector unit's §18.5. (ii) $\Longleftrightarrow$ (iii) by §12.2. (ii) gives (iv) because two independent vectors span $\mathbb{R}^2$ (§18.5 again), and (iv) gives (iii) since $\dim \mathbb{R}^2 = 2$. (iv) $\Longleftrightarrow$ (v) is the solvability theorem of §10.3. □

When $\det(\mathbf{A}) = 0$ the rank is 1 or 0, and it is 0 only for the zero matrix. One table holds the whole picture.

$\operatorname{rank}(\mathbf{A})$	$\det(\mathbf{A})$	$\operatorname{col}[\mathbf{A}]$	what $\mathbf{A}$ does to the plane
2	$\neq 0$	$\mathbb{R}^2$	moves it onto all of itself
1	0	a line through $\mathbf{0}$	flattens it onto that line
0	0	$\{\mathbf{0}\}$	crushes it to the origin

14.3. The size of the determinant: area

In §13.4 the unit square, of area 1, became the parallelogram with sides $\mathbf{a}_1$ and $\mathbf{a}_2$. The determinant measures that parallelogram:

$$\text{area of the parallelogram with sides } \mathbf{a}_1, \mathbf{a}_2 \;=\; |\det(\mathbf{A})|.$$

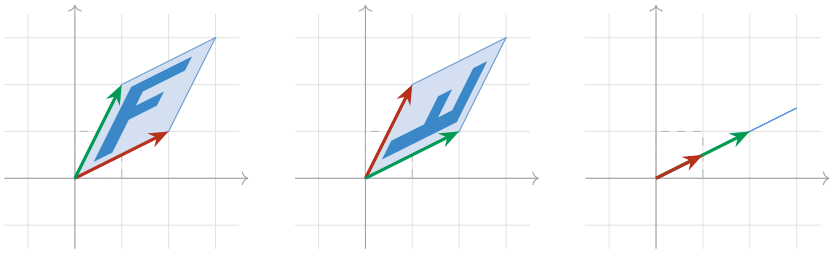
This is stated here and derived in §23 from base times height. For $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ of §13.4, the parallelogram with corners $\mathbf{0}$, $[2,1]$, $[3,3]$, $[1,2]$ has area 3, and $\det(\mathbf{A}) = 3$.

And it is not only the unit square. Every square of the grid is carried to a copy of the same parallelogram, so every region of the plane has its area multiplied by the same factor $|\det(\mathbf{A})|$. The determinant is the *area scale factor* of the matrix.

14.4. The sign of the determinant: orientation

The area is $|\det(\mathbf{A})|$; the sign carries separate information. Turning from $\mathbf{e}_1$ to $\mathbf{e}_2$ is a quarter-turn counterclockwise.

- If $\det(\mathbf{A}) > 0$, turning from $\mathbf{a}_1$ to $\mathbf{a}_2$ (the short way) is also counterclockwise. The plane is moved without being turned over.
- If $\det(\mathbf{A}) < 0$, turning from $\mathbf{a}_1$ to $\mathbf{a}_2$ is clockwise. The plane has been flipped, as in a mirror, and the letter F comes out reversed.



$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $\det = 3$ $\quad$ $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, $\det = -3$ $\quad$ $\begin{bmatrix} 1 & 2 \\ 0.5 & 1 \end{bmatrix}$, $\det = 0$

In each picture the red arrow is $\mathbf{A}\mathbf{e}_1$ and the green arrow is $\mathbf{A}\mathbf{e}_2$. On the left, area 3 and the F reads correctly. In the middle, the columns of the left matrix have been swapped: the same parallelogram, area 3 again, but the F is mirrored and the determinant is -3. On the right the columns are parallel and the square is flattened into a segment.

14.5. When the determinant is zero

A zero determinant means the parallelogram has no area: its sides are parallel, and the plane is flattened onto the line col[$\mathbf{A}$]. Something is lost in the flattening, and it can be located exactly.

Example 14.2. Take $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ again, with $\det(\mathbf{A}) = 0$ and column space the line through $[\,1, 2\,]$. Look at the vector $[\,2, -1\,]$:

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2-2 \\ 4-4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

By linearity every multiple $t[\,2, -1\,]$ is sent to $\mathbf{0}$ too. A whole line of the plane is crushed to a single point. And because of that, different inputs can give the same output: $\mathbf{Ax} = \mathbf{Ay}$ whenever $\mathbf{x} - \mathbf{y}$ lies on that line. Once the matrix has acted, there is no way to tell those inputs apart.

That line, the set of vectors a matrix sends to $\mathbf{0}$, is the *null space*. It is the subject of Part 6, where it settles the second half of the question §10.3 left open: when a system has solutions, whether it has one or infinitely many.

14.6. Practice

P1. Describe col[$\mathbf{A}$] and row[$\mathbf{A}$] for $\mathbf{A} = \begin{bmatrix} 1 & -1 & 2 \\ -2 & 2 & -4 \end{bmatrix}$, and give rank($\mathbf{A}$).

P2. For which $\mathbf{b}$ does $\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \mathbf{x} = \mathbf{b}$ have a solution? Is $\mathbf{b} = [\,4, 8\,]$ one of them? Is $\mathbf{b} = [\,4, 7\,]$?

P3. A linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^3$ has $T(\mathbf{e}_1) = [\,1, 0, 2\,]$ and $T(\mathbf{e}_2) = [\,-1, 3, 0\,]$. Write its matrix and compute $T([\,2, 1\,])$.

P4. Compute the determinant of $\begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$, $\begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$ and $\begin{bmatrix} 6 & -2 \\ -9 & 3 \end{bmatrix}$. For each, give the rank and the area of the image of the unit square, and say whether the plane is flipped.

P5. Find a nonzero vector that $\begin{bmatrix} 6 & -2 \\ -9 & 3 \end{bmatrix}$ sends to $\mathbf{0}$.

Answers

P1. The columns $[\,1, -2\,]$, $[\,-1, 2\,]$, $[\,2, -4\,]$ are all multiples of $[\,1, -2\,]$, so col[$\mathbf{A}$] is the line through $[\,1, -2\,]$ in $\mathbb{R}^2$ and rank($\mathbf{A}$) = 1. The second row is -2 times the first, so row[$\mathbf{A}$] is the line through $[\,1, -1, 2\,]$ in $\mathbb{R}^3$, of dimension 1, as §12.4 requires.

P2. The column space is the line through $[\,1, 2\,]$ (the second column is 3 times the first), i.e. the vectors $[\,b_1, b_2\,]$ with $b_2 = 2b_1$. So $[\,4, 8\,]$ is one of them (solutions exist, e.g. $\mathbf{x} = [\,4, 0\,]$) and $[\,4, 7\,]$ is not.

P3. The columns are the images of $\mathbf{e}_1$ and $\mathbf{e}_2$: $\begin{bmatrix} 1 & -1 \\ 0 & 3 \\ 2 & 0 \end{bmatrix}$, and $T([\,2, 1\,]) = 2[\,1, 0, 2\,] + 1[\,-1, 3, 0\,] = [\,1, 3, 4\,]$.

P4. $\det = 6 - 4 = 2$: rank 2, area 2, not flipped. $\det = -2$: rank 2, area 2, flipped (the x-direction is reversed). $\det = 18 - 18 = 0$: the second column is $-\frac{1}{3}$ times the first; rank 1, area 0; the square is flattened, so the question of flipping does not arise.

P5. The rows are both multiples of $[3, -1]$ ($6x - 2y = 0$ and $-9x + 3y = 0$ are the same equation $y = 3x$), so $[1, 3]$ works: $6 - 6 = 0$ and $-9 + 9 = 0$.

Looking ahead. Part 3 uses §13.3 as a construction kit. It takes the motions of the plane one meets in geometry — scalings, shears, rotations, reflections, projections, and one motion that turns out not to be linear at all — and builds each one's matrix from where it sends $\mathbf{e}_1$ and $\mathbf{e}_2$. Then it asks which matrices move the plane without changing any length, and finds that the answer is a single matrix equation.